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DSP based real-time single carrier underwater acoustic communications using frequency domain turbo equalization



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ABSTRACT

In this paper, a DSP based real-time single carrier underwater acoustic communication system is developed. The designed receiver system operates in frequency domain for low complexity equalization. The turbo principle is incorporated into channel equalization and channel estimation processes to improve the system reliability. The channel is treated as stable within each small block and iteratively estimated based on LMS method and soft information provided by channel decoder. The soft LMS channel estimation is able to achieve better performance than conventional phase compensation scheme in single carrier frequency domain system. The receiver system is implemented on a floating point DSP running at 350 MHz. Through careful optimization, the real-time processing is achieved with sufficient margin for both single-hydrophone receiver and multi-hydrophone systems. The designed system has been tested by both tank experiment and sea trials and achieves satisfied performance.

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1. Introduction

In recent years, there has been an increasing interest in the development of high performance underwater acoustic communication networks for various underwater applications. One of the key issues for the success of the underwater acoustic networks is to build up a reliable and high performance physical underwater acoustic communication link.

To develop reliable and high performance underwater acoustic communication link, various techniques have been proposed and investigated [1]. Recently, orthogonal frequency-division multiplexing (OFDM) and single carrier frequency domain equalization (SC-FDE) have been extensively studied for underwater communication due to their low computational complexity. OFDM transmission with two-stage Doppler compensation has been proposed

for underwater communication for single-input multiple-output (SIMO) [2] and multiple-input multiple-output (MIMO) [3] situations. Compared with OFDM transmission, SC-FDE exhibits the property of low peak to average power ratio due to its single carrier transmission nature while the low complexity frequency domain equalization could reduce the complexity of the system. The investigation of SC-FDE for underwater acoustic communication has been carried out in [4] for SISO situation and in [5] for MIMO situation.

To further improve the reliability of underwater communication, iterative turbo detection technique where the extrinsic information from channel decoder is feedback to the equalizer has been widely investigated in underwater acoustic communication applications [6–9]. The adaptive time domain turbo equalization for SISO and MIMO underwater communication was proposed in [6]. In [7], a MIMO time domain turbo equalizer with hybrid soft interference cancellation and reliability ordering was proposed for single carrier communication. A frequency domain turbo equalizer with successive interference cancellation and phase correction was investigated in [8] for

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single carrier MIMO transmission. Iterative receivers for distributed MIMO OFDM were investigated in [9].

The above mentioned works are carried out in the view of underwater acoustic communication research and real-time implementation is not considered, which makes those works still far away from practical underwater acoustic communication system. Some work has been done on the development of the underwater acoustic modem [10–16]. The designs in commercial products [10–12] are either based on non-coherent FSK detection [10,12] or direct spread spectrum technique [11]. All of them operate in a non iterative manner and inherently have low band efficiency. [16] considered the design of low cost modem and the system was also based on FSK. The Micro-Modem in [13] can operate in two modes: (1) low data rate mode based on non-coherent FSK detection and (2) high data rate mode based on PSK. But the time domain decision feedback equalizer is used and operates in non iterative way which may suffer performance degradation due to error propagation. [14] investigates the implementation of DSP based SISO and MIMO OFDM underwater modem and the receiver also operates non-iteratively. The DSP implementation of turbo equalizer has been studied in [17]. The equalizer operates in time domain while the taps of equalizer are calculated in frequency domain. However, the channel state information is assumed to be known and not updated during iteration process. The underwater acoustic modem with turbo equalizer is also investigated in [15]. However, the time domain equalizer inherently exhibits higher computational complexity than its frequency domain counterpart.

Our work focus on the issue of real-time implementation of frequency domain turbo equalizer for underwater acoustic communication. We design a real-time underwater acoustic communication receiver based on a digital signal processor (DSP). In this paper, we present our receiver structure and DSP implementation issues. To verify the system performance, both tank and sea experiments are carried out. Experiments show that we can achieve real-time processing. The BERs achieved in the experiments are below 5×10^{-3} for single-hydrophone system and less than 10^{-3} for two-hydrophone system.

The rest of the paper is organized as follows. The transmitter scheme and receiver system are introduced in Sections 2 and 3 respectively. The DSP implementation issues are introduced in Section 4 and Section 5 presents the results and analysis. Finally, Section 6 gives some conclusions.

Notation. Upper-case (resp. lower-case) bold letters represent matrices (resp. column vectors) with $[A]_{n,m}$ (resp. a_n) denoting the (n, m) th (resp. n th) entry of $\mathbf{A}$ (resp. $\mathbf{a}$); $(\cdot)^T$, $\text{conj}(\cdot)$ and $(\cdot)^H$ denote transpose, conjugate and Hermitian transpose operators, respectively. $\mathbf{F}_N$ and $\mathbf{I}_N$ represent N by N normalized FFT matrix and N by N unit matrix respectively. $\mathbf{e}_i$ is a column vector with i th entrance being 1 and other entrances being 0 and $\mathbf{1}_G$ is a G by 1 column vector with all the entrances being 1. $\hat{\mathbf{a}}$ denotes the FFT of $\mathbf{a}$, $\hat{a}$ represents the estimation of a , $\hat{a}$ is the expectation of a , $\text{tri}(\mathbf{A})$ means the trace of matrix $\mathbf{A}$ and $\text{diag}(\mathbf{A})$ denotes the diagonal of matrix $\mathbf{A}$. $\otimes$ and $\odot$ denote Kronecker product and Hadamard product.

Table 1
Symbol mapping.

00	01	10	11
$\frac{-1-i}{\sqrt{2}}$	$\frac{-1+i}{\sqrt{2}}$	$\frac{1-i}{\sqrt{2}}$	$\frac{1+i}{\sqrt{2}}$

2. Transmitter

The information bits a_k are divided into blocks with block length N , then the information bits in each block are encoded by a rate $\frac{1}{2}$ convolutional code with the generation polynomial as $[121/37]_8$. The coded bits $c_k[m]$ are interleaved by a random interleaver, and then mapped to QPSK symbols $u_k[m]$ with respect to the given constellation set $\mathbb{S} = \{\alpha_q\}_{q=1}^4$ as in Table 1 for the m th block. A PN sequence v_k is cascaded in the tail of each block as unique words (UW). The length of UW is enough long to avoid inter block interference (IBI). Therefore, the transmitted baseband signals for the m th data block can be denoted as $\mathbf{s}[m] = [\mathbf{u}[m]^T \mathbf{v}^T]^T$. The generated transmitted baseband signals are pulse shaped by a raised cosine filter and upconverted to passband. The diagram of transmitter is shown in Fig. 1.

3. Receiver

At receiver, the received passband signal is first downconverted to baseband. In practical communication, the received signal usually suffers carrier frequency offset (CFO) due to Doppler effect and clock drifting between transmitter and receiver. To compensate CFO, three identical pilot data blocks with block duration of N_c symbols are transmitted. The choice of N_c should be larger than the channel delay spread. Therefore, CFO f_d is estimated as [18]

$$\hat{f}_d = \frac{1}{\pi N_c} \sum_{i=N_c}^{2N_c-1} \arg \{x[i] \text{conj}(x[i + N_c])\}, \quad (1)$$

where $x[i]$ is the received signal corresponding to the CFO pilot. After CFO compensation and with the quasi-static channel assumption i.e. channel is stable within every receiving block, the CFO compensated time-domain baseband signals within k th receiving block are written as

$$\mathbf{y}[k] = \mathbf{H}[k]\mathbf{s}[k] + \mathbf{w}[k] \quad (2)$$

where $\mathbf{y}[k]$ is composed of the CFO compensated baseband signals in the k th block and $\mathbf{H}[k]$ is a circulant matrix with the first column as

$$[h[0; k] \cdots h[L-1; k] \mathbf{0} \cdots \mathbf{0}]^T \quad (3)$$

where $h[i; k]$ represents the baseband channel impulse response (CIR) in the k th block and $\mathbf{w}[k]$ is the zero mean Gaussian white noise with variance $\sigma_0^2[k]$. $\mathbf{s}[k]$ is composed of transmitted baseband signals considering the oversampling factor G . More specifically, $\underline{s}_i[k]$ is defined as

$$\underline{s}_i[k] = \begin{cases} s_{\frac{i}{G}}[k] & \text{if } \frac{i}{G} \text{ is integer} \\ 0 & \text{else.} \end{cases} \quad (4)$$

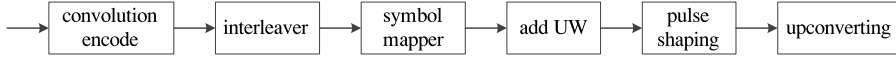


Fig. 1. Diagram of transmitter.

Applying FFT on (2), we get the frequency domain model as

$$\begin{aligned}\tilde{\mathbf{y}}[k] &= \tilde{\mathbf{H}}[k]\tilde{\mathbf{s}}[k] + \tilde{\mathbf{w}}[k] \\ &= \tilde{\mathbf{H}}[k]\tilde{\mathbf{s}}[k] + \tilde{\mathbf{w}}[k],\end{aligned}\quad (5)$$

where $\tilde{\mathbf{H}}[k]$ is a diagonal matrix with $\text{diag}(\tilde{\mathbf{H}}[k]) = \sqrt{GN}\mathbf{F}_{GN}\mathbf{H}[k]\mathbf{e}_1$ and $\tilde{\mathbf{H}}[k] = \mathbf{H}[k](\mathbf{1}_G \otimes \mathbf{I}_N)$.

3.1. Turbo equalization

The frequency domain turbo equalizer is based on [19]. Instead of using a symbol rate equalizer as [19], the oversampling factor $G = 2$ is adopted in our work. The estimation of $\tilde{\mathbf{s}}$ can be written as

$$\hat{\tilde{\mathbf{s}}}[k] = \tilde{\mathbf{s}}[k] + \mathbf{\Omega}[k](\tilde{\mathbf{y}}[k] - \tilde{\mathbf{H}}[k]\tilde{\mathbf{s}}[k]), \quad (6)$$

where $\mathbf{\Omega}[k]$ represents a column vector composed of equalizer coefficients. Based on linear maximum mean square error, it can be computed as

$$\mathbf{\Omega}[k] = \left(\frac{\sigma_0^2[k]}{\sigma_s^2[k]} \mathbf{I}_N + \tilde{\mathbf{H}}[k]^H \tilde{\mathbf{H}}[k] \right)^{-1} \tilde{\mathbf{H}}[k]^H, \quad (7)$$

where $\sigma_s^2[k]$ is the mean of symbol variance in block k . The required symbol mean and symbol variance are calculated by employing the extrinsic log likelihood ratios (LLR) from channel decoder. The detailed process can refer to [20]. The obtained frequency-domain symbol estimations are transferred back to time domain by FFT operation.

3.2. Soft-demapper

The soft-demapper is adopted to computed extrinsic LLRs needed by channel decoder. To calculate the extrinsic LLRs, the time-domain symbol estimation is approximated as

$$\hat{s}_i[k] = \bar{s}_i[k] + \kappa[k](s_i[k] - \bar{s}_i[k]) + v[k], \quad (8)$$

where $v[k]$ is a zero-mean Gaussian noise with variance $\sigma_{e_i}^2[k]$ and $\kappa[k] = \frac{1}{N} \text{tri}(\mathbf{\Omega}[k]\tilde{\mathbf{H}}[k])$. The noise variance $\sigma_{e_i}^2[k]$ is calculated as

$$\begin{aligned}\sigma_{e_i}^2[k] &\approx \frac{\sigma_0^2}{N} \text{tri}(\mathbf{\Omega}[k]\mathbf{\Omega}[k]^H) \\ &\quad + \frac{\sigma_{s_i}^2[k]}{N} \text{tri}(\mathbf{\Omega}[k]\tilde{\mathbf{H}}[k]\tilde{\mathbf{H}}[k]^H\mathbf{\Omega}[k]^H),\end{aligned}\quad (9)$$

where $\sigma_{s_i}^2[k] = \frac{1}{N} \sum_{i=0}^{N-1} \sigma_{s_i}^2[k] - \frac{\sigma_{s_i}^2}{N}$ in which $\sigma_{s_i}^2$ denotes the variance of s_i . To simplify the calculation of extrinsic LLRs, we approximate the extrinsic LLR as

$$LLR_e(c_{k,i}) = \ln \frac{\sum_{s_k \in \hat{S}_i^1} P(\hat{S}_k | s_k)}{\sum_{s_k \in \hat{S}_i^0} P(\hat{S}_k | s_k)}, \quad (10)$$

where $\hat{S}_i^d \triangleq \{\alpha_q | \alpha_q \in \mathbb{S} : c_{q,i} = d\}$ with $d \in \{0, 1\}$. $\{c_{q,i}\}_{i=1}^2$ are the demapping bits of the symbol α_q and $P(\hat{S}_k | s_k)$ denotes the conditional probability of $\hat{S}_k$.

Considering the constellation set in Table 1, we have

$$LLR_e(c_{i,1}) = \frac{2\sqrt{2}\kappa[k]\text{Re}(\hat{s}_i[k] - \bar{s}_i[k] + \kappa[k]\bar{s}_i[k])}{\sigma_{e_i}^2[k]}, \quad (11)$$

and

$$LLR_e(c_{i,2}) = \frac{2\sqrt{2}\kappa[k]\text{Im}(\hat{s}_i[k] - \bar{s}_i[k] + \kappa[k]\bar{s}_i[k])}{\sigma_{e_i}^2[k]} \quad (12)$$

corresponding to $\hat{s}_i[k]$.

3.3. Channel estimation

For practical receiver, the channel matrix $\mathbf{H}[k]$ needs to be estimated. In the proposed receiver, the channel estimation task is divided into two phases: (1) initial estimation and (2) channel tracking.

The initial channel estimation is realized by pilots based time-domain least square estimation. More specifically, the transmitted data blocks are grouped into several frames. At the beginning of each frame, a group of pilots $t[n]$ with length of N_p are inserted. At the receiver side, the corresponding CFO compensated baseband signals can be expressed as

$$\mathbf{z} = \mathbf{T}\mathbf{h} + \mathbf{w}, \quad (13)$$

where $\mathbf{z} = [z[L-1], \dots, z[N_p-1]]^T$ is a column vector composed of the corresponding CFO compensated baseband signals, $\mathbf{h} = [h[0; 0] \dots h[L-1; 0]]$ is the related CIR and $\mathbf{w}$ is the noise vector. Matrix $\mathbf{T}$ is composed of pilots and defined as

$$\mathbf{T} \triangleq \begin{bmatrix} \underline{t}[L-1] & \dots & \underline{t}[1] & \underline{t}[0] \\ \underline{t}[L] & \dots & \underline{t}[2] & \underline{t}[1] \\ \vdots & \ddots & \ddots & \vdots \\ \underline{t}[N_p-1] & \dots & \underline{t}[N_p-L+1] & \underline{t}[N_p-L] \end{bmatrix} \quad (14)$$

where $\underline{t}[n]$ is defined as

$$\underline{t}[n] = \begin{cases} t\left[\frac{n}{G}\right] & \text{if } \frac{n}{G} \text{ is integer} \\ 0 & \text{else.} \end{cases} \quad (15)$$

Thus, the estimation of $\mathbf{h}$ can be computed as $\hat{\mathbf{h}} = (\mathbf{T}^H \mathbf{T})^{-1} \mathbf{T}^H \mathbf{z}$. With $\hat{\mathbf{h}}$, $\hat{\mathbf{H}}[k]$ can be calculated for equalizer.

For channel tracking, we use adaptive iterative channel tracking. The symbol mean generated by extrinsic LLRs from channel decoder is adopted as training symbol. Based on least mean square criteria (LMS), the estimation of channel matrix $\hat{\mathbf{H}}[k]$ is updated as [21]

$$\begin{aligned}\text{diag}(\hat{\mathbf{H}}[k]) &= \text{diag}(\hat{\mathbf{H}}^{j-1}[k]) + \mu \mathbf{F}_{2N}^H \left(\text{conj}(\tilde{\mathbf{s}}^{j-1}[k]) \right. \\ &\quad \left. \odot (\tilde{\mathbf{y}}[k] - \hat{\mathbf{H}}^{j-1}[k]\tilde{\mathbf{s}}^{j-1}[k]) \right)\end{aligned}\quad (16)$$

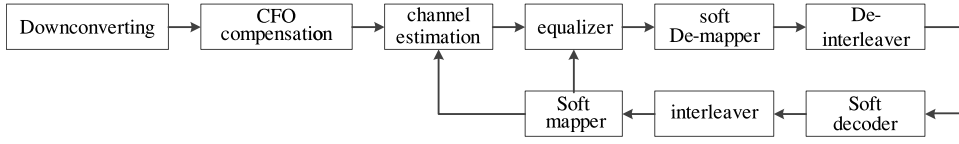


Fig. 2. Receiver structure with soft channel estimation.

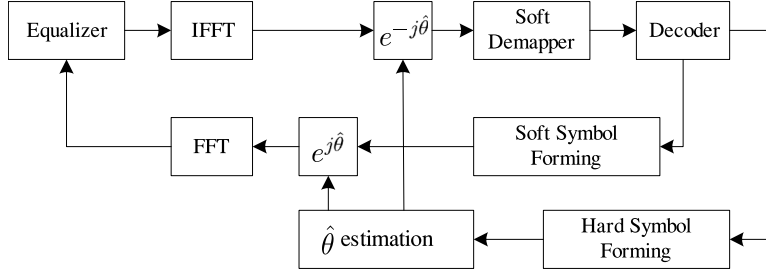


Fig. 3. Iterative phase compensation.

where $\hat{\mathbf{H}}^j[k]$ is the estimation of $\tilde{\mathbf{H}}[k]$ in the j th iteration and $\bar{\mathbf{S}}^j[k]$ denotes the symbol mean in the j th iteration. $\hat{\mathbf{H}}^j[k]$ is then converted to $\tilde{\mathbf{H}}^j[k]$ which is used to replace $\tilde{\mathbf{H}}[k]$ in turbo equalizer and soft-demapper. The whole receiver structure with iterative channel tracking is shown in Fig. 2.

For the first iteration in block k , the obtained $\tilde{\mathbf{H}}^j[k-1]$ in the $(k-1)$ th block is used where J is the pre-determined maximum iterations for one data block.

Another method for channel tracking is based on the work in [4] which incorporates a separate phase rotation compensation scheme. Enlightened by work in [22], we extend the method in [4] to an iterative manner. In this method, the time-domain phase-uncompensated symbol estimation in the m th iteration $\hat{s}_{i,rot}^m[k]$ is approximated as

$$\hat{s}_{i,rot}^m[k] = \bar{s}_i^m[k] e^{j\theta_i^{m-1}[k]} + \kappa[k] (s_i[k] e^{j\phi_i}[k] - \bar{s}_i^m[k] e^{j\theta_i^{m-1}[k]}) \quad (17)$$

where ϕ_i is an unknown phase rotation and $\theta_i^{m-1}[k]$ is the estimated phase rotation in $(m-1)$ th iteration. Instead of using the tentative decision as [4] to estimate the rotated phase, we use the decoder's hard decision in the last iteration. After phase rotation compensation, we approximate $\hat{s}_i^m[k]$ as (8) to calculate extrinsic LLRs. The structure for the iterative phase compensation scheme is shown in Fig. 3. For the initial channel estimation in the k th block, we use the frequency domain LMS adaptive channel estimation as [23] where the hard decision made by decoder in the $(k-1)$ th block is treated as training symbol.

4. DSP implementation

The system is implemented on a DSP EVM board as shown in Fig. 4. The core processor is a floating-point DSP TMS320C6727B which operates at 350 MHz with 256 kB on chip RAM. It can execute up to eight instructions simultaneously every CPU cycle. The DSP processor also provides a



Fig. 4. Evaluation board.

plenty of peripherals which facilitate the system development. It is equipped with one dual data movement accelerator (dMAX) controller, three multichannel audio serial ports (McASPs) and one external memory interface (EMIF) etc. The EVM board also comprises multichannel analog-to-digital converters (ADCs) and external synchronous dynamic random access memory (SDRAM) up to 128 MB. In our development, one serial port McASP0 is configured to support integrated interchip sound protocol for fetching audio data from ADC. dMAX controller transports the data in the receive buffer of McASP0 to the external SDRAM. In this way, CPU is released from the data movement between memory and the device peripherals and focused on data processing. The data flow for the received signals is shown in Fig. 5.

Two data buffers with same size are allocated in the external SDRAM and the incoming data are stored into these two buffers alternatively. For example, CPU processes data stored in buffer 1 while new data are stored into buffer 2. When buffer 1 is full, dMAX controller will generate a CPU interrupt. And the roles of buffer 1 and buffer 2 alternate. Therefore, CPU has to finish processing the current data buffer before the other buffer is full.

The system is implemented by C language to speed up system development and favor code compatibility.

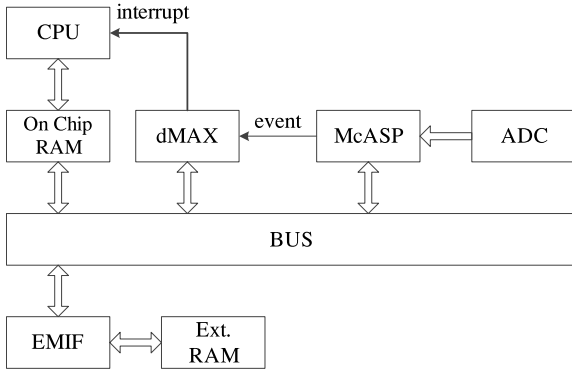


Fig. 5. Data flow of received signals.

In addition, the C code is compiled by Code Composer Studio with optimization level 3. The system also takes the advantage of FFT from TI Digital Signal Processing Library and TI C67x MATHLIB. We also have applied several coding strategies to reduce program executing time. For example, the 32-bit floating point instead of 64-bit floating point is employed to represent data for the purpose of shortening executing time and saving on-chip memory. Besides, we also use loop unrolling to accelerate program execution [24].

In addition to adopt the appropriate coding strategies, we also apply proper algorithms for some heavy-computational tasks. The FFT and overlap-add method [25] are utilized for linear convolution.

For the pilot-based time-domain channel estimation, we pre-calculate the pseudo-inverse matrix and store the matrix entries in the memory. Due to the relatively large dimension of the matrix and the limited on-chip memory, the pre-computed matrix has to be stored in external memory. Since the EMIF port of TMS320C6727B DSP can only support SDRAM with maximum clock rate at 133 MHz and DSP does not contain any data cache and unified L2 cache, there will be a bottleneck when CPU requires large amounts of data from external memory. However, we manage to release this bottleneck through several strategies:

- (1) The m-sequence is employed as training sequence which means the entries in the resulting matrix would be real number. Comparing with complex-number matrix, we can reduce the overhead of accessing external memory. Besides, we can reduce one complex-number multiplication into two real-number multiplications hence reducing execution time.
- (2) Although the oversampling technique with oversampling factor $g = 2$ is employed, it should be noted that the pseudo-inverse matrix $\mathbf{P}_{over.} = (\mathbf{T}^H \mathbf{T})^{-1} \mathbf{T}^H$ in our system can be written as $\mathbf{P}_{over.} = \mathbf{P}_{sym.} \otimes \mathbf{I}_2$ where $\mathbf{P}_{sym.}$ corresponds to the pseudo-inverse matrix in the symbol-rate sampled case. Hence, we can set a buffer with the same size as the row of matrix $\mathbf{P}_{sym.}$ in on-chip memory and load every row in $\mathbf{P}_{sym.}$ into the allocated buffer. After each loading, we can calculate the two corresponding coefficients in the estimated channel impulse response. In this way, we further reduce the overhead of external-memory access and shorten execution time.

Table 2

System parameters.

Carrier frequency f_c	12 kHz
Sampling frequency f_s	48 kHz
Block duration T_b	128 ms
Number of symbols per block N_b	256
Number of UW symbols per block $N_{t,b}$	50
LMS step size μ	0.001

The system operates in two phases: (i) synchronization and (ii) communication. At the beginning of each burst data transmission, the chirp signal is transmitted due to its good correlation property. At receiver side, the matched filter is used to detect the chirp signal. When the ratio between the current output of the matched filter and the average matched-filter output within a sliding window exceeds the threshold, the chirp signal is assumed to be detected. After detecting the significant peak, the system switches into communication phase.

5. Experimental results and analysis

To evaluate the system performance, we carried out experiments both in tank and sea. The transmitting signals are generated by using Matlab on PC and stored in WAV format. During experiments, those signals are loaded into SOUND DEVICE 722 for broadcasting. The frame structure of transmitted signals is presented in Fig. 6, and the system parameters are listed in Table 2.

In current experiment setting, each frame contains 6 data blocks resulting in an effect data rate at 770 bits/s in experiment. However, it is worth to note that current experiment setting is quite conservative. In practical implementation, the averaging window length for LS channel estimation is chosen to be 150 ms, which means the pilot duration in our setting is much longer than required. Moreover, parts of CFO signal can also be used for LS channel estimation. Therefore, the pilot duration can be further reduced in future system realization and the effect data rate can be thereby increased. Besides, the maximum delay spread in our setting is much larger than the delay-spread values reported in many other works [4,2,3], which means the maximum delay spread might be reduced with respect to practical environments.

The required CPU cycles for different process procedures are listed in Table 3 where Equalizer 1 represents the normal frequency domain turbo equalizer while Equalizer 2 represents the frequency domain turbo equalizer with phase correction. For underwater acoustic communication, a receiver array with multiple hydrophones is often used to obtain receiving diversity. Thus, the processing time for receiving system with multi-channel equalizer (2 channels) is also included in Table 3. Table 4 presents the CPU processing for one received block. For data block, we set the maximum iteration to be 4. We only present the receiver with adaptive iterative channel tracking in Table 4, since the receiver with adaptive iterative channel tracking achieves better performance than the receiver with iterative phase compensation in our experiments. As it is shown in Table 4, there is still large margin on the execution time compared with our block duration i.e. 128 ms, which means higher symbol rate can also be further implemented with current system.

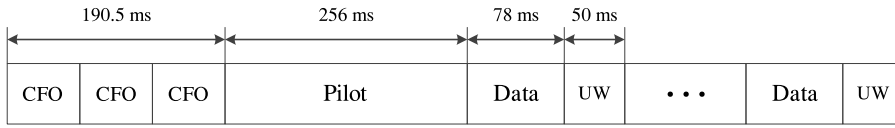


Fig. 6. Frame structure.

Table 3
CPU cycles for different functions.

Function	CPU cycles
MAP decoder	1,619,061 (4.63 ms)
Downconverting	2,813,424 (8.1 ms)
LS channel estimation	3,539,245 (10.1 ms)
Soft mapper	147,262 (0.42 ms)
Single equalizer 1	232,808 (0.67 ms)
Single equalizer 2	282,849 (0.81 ms)
Soft LMS channel estimation	198,000 (0.566 ms)
Phase estimation	63,752 (0.18 ms)
Two-channel equalizer	258,631 (0.74 ms)
Two-channel downconverting	5,905,541 (16.9 ms)
Two-channel LS estimation	7,080,835 (20.2 ms)

Table 4
CPU cycles per data block.

Type	CPU time (ms)
Single channel pilot block	19
Single channel data block	29
Two-channel pilot block	38
Two-channel data block	38

5.1. Tank experiment

The tank experiment was carried out in the Acoustic Underwater Lab at Norwegian University of Science and Technology (NTNU) with a single transducer and a single receiving hydrophone. The tank is 4 m long, 3 m wide and 2 m deep. The transducer and hydrophone were both deployed at the depth 1 m with the horizontal distance 1 m. The measured CIR is presented in Fig. 7. The delay spread in the tank was almost 50 ms, and we noticed the CFO due to clock drifting between SOUND DEVICE and DSP EVM board. The bit error rate performances for both iterative phase compensation and soft LMS method are shown in Fig. 8. We found that the receiver with soft LMS method outperforms the receiver with iterative phase compensation. This can be ascribed to the reason that soft decision is more robust compared with hard decision. And the estimated phase rotation is much more easily distracted by residual ISI compared with soft iterative channel estimation.

5.2. Sea trial 1

The sea experiment was conducted at a guest pier, on April 27, 2014, Trondheim, Norway. The transmitted signal parameters were the same as those in the tank experiment. The water depth was around 12 m and the transmission range in the experiment was around 30 m. The transmitting transducer was deployed at the depth about 12 m, while at the receiver side a vertical array with 4 hydrophones was deployed. The vertical hydrophone array covered the water column from 5 to 6.5 m with a spatial spacing of 0.5 m. During the experiment, each

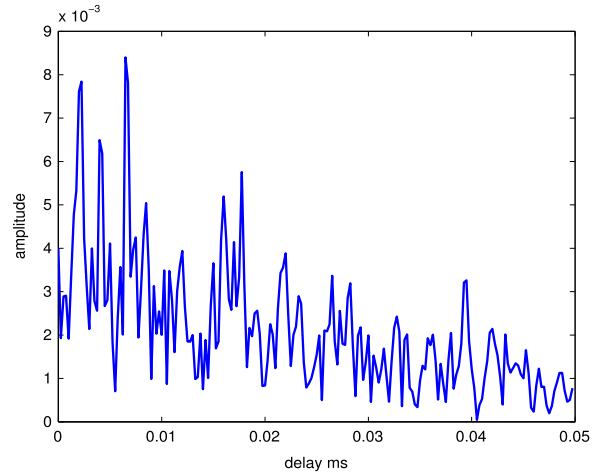


Fig. 7. Measured CIR in tank.

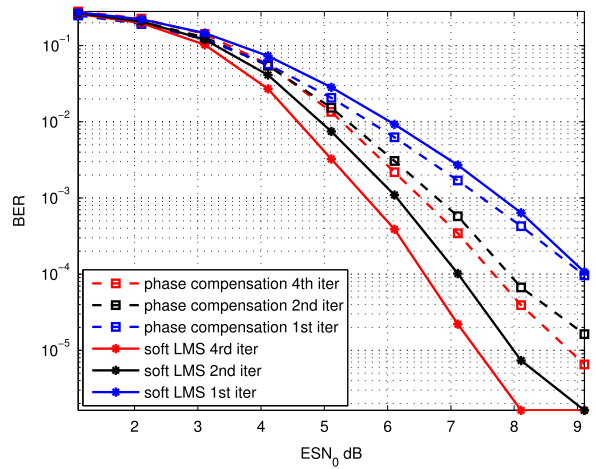


Fig. 8. BER performance of tank experiment.

hydrophone was connected with DSP board for on-line processing alternatively. We used phase compensation method during on-line processing but the received data were stored on the external RAM of EVM board for future processing. Both the transducer and the hydrophone array were drifting in the water. Therefore their positions can easily be affected by currents. One measured CIR evolution as time is presented in Fig. 9. There are mainly three arrival paths with delay spread about 5 ms during the experiment. And the second arrival path exhibits much time variability. During the experiments, 20 burst transmissions were carried out. In total 20 bursts, the soft LMS method achieved error free after 4 iterations, while the iterative phase compensation method achieved error free in 11 bursts after the same number of iterations. The worst BERs

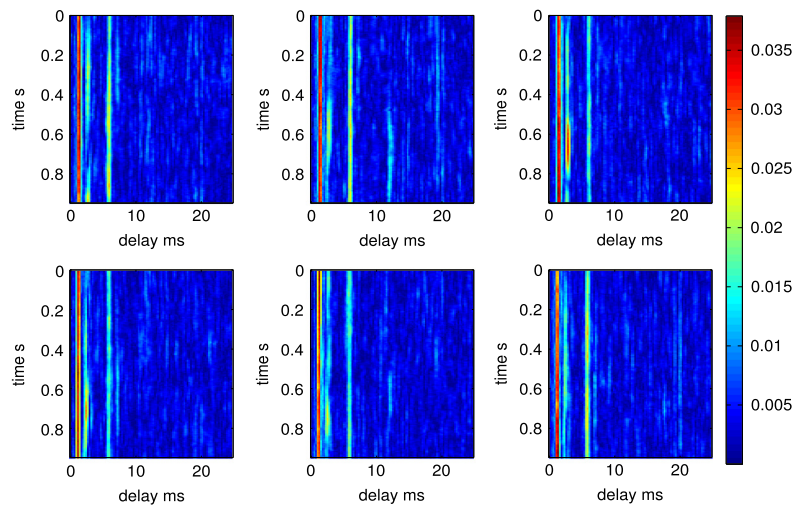


Fig. 9. Typical CIR during sea trial.

Table 5
BER in sea trial 1.

Receiver structure	Hydrophone Nr.	ITER. 1	ITER. 2	ITER. 4
Iterative phase	2	13.73%	12.67%	10.39%
	3	0.78%	0.23%	0.23%
	4	7.86%	6.98%	5.71%
Soft LMS	2	4.5%	0.51%	0
	3	0	0	0
	4	1.39%	0	0

achieved for hydrophone 2–4 are presented in Table 5. When hydrophone 1 is used for receiving, both methods achieve error free transmission. This was because the sea was relatively calm during that period. To further illustrate the system performance, we also generate the BER curve for the channel state in Fig. 10. To produce the BER curve, we add the synthetic Gaussian noise into the collected data set. The sea trial results also show that the soft LMS method outperforms the iterative phase compensation method. During all data bursts, the soft LMS method can achieve error free transmission while the phase compensation method would fail when the channel exhibits relatively high time variability.

5.3. Sea trial 2

This experiment was also carried out in the Trondheim fjord, July 2, 2014. At the beginning of the experiment, a 8-element vertical array with 1 m spatial interval was deployed from NTNU's research vessel R/V Gunnerus, and the transducer was close shore deployed. The experiment setting is shown in Table 6. We have tried to go bit further away from shore for instance at 710 m, but failed to receive any signal. It might be due to the fact that the receiver array was within the shadow zone. During this experiment, the communication data were collected by a data acquisition device and later processed by the EVM board at the lab. Since the soft LMS method achieves better performance

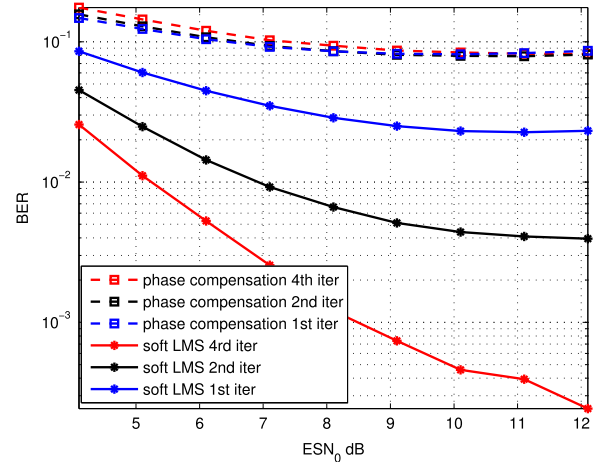


Fig. 10. BER performance of typical channel in sea trial 1.

Table 6
Experiment setup in sea trial 2.

Horizontal range	100 m
Array depth	20–28 m
Transducer depth	6 m
UW length	50 ms

during previous experiments, we only considered the soft LMS method to update the channel information in this sea trial. We also implemented the multi-channel equalizer to explore the spatial gain. With respect to the physical size of the practical modem, the two adjacent hydrophones were employed for multi-channel processing. The BER performances of this sea trial was presented in Table 7. The worst BER achieved by single hydrophone receiver was on the order of 10^{-3} , and multi-channel receiver could significantly improve the system performance, achieving error free in the test by obtaining spatial gain.

Table 7

BER performance during transmission from shore to vessel in sea trial 2.

Setup	Iter. 1	Iter. 2	Iter. 4
Average single hydrophone	0.0396%	0.00713%	0.00713%
Worst single hydrophone	0.15%	0.11%	0.11%
Two hydrophones	0	0	0

Table 8

BER performance during sea trial 3.

Setup	Iter. 1	Iter. 2	Iter. 4
Average single hydrophone	2.4%	0.66%	0.04%
Worst single hydrophone	12.41%	4.86%	0.41%
Average two hydrophones	0.52%	0.057%	0.0047%
Worst two hydrophones	0.75%	0.0326%	0.0326%

5.4. Sea trial 3

The 3rd sea trial was also carried out in Trondheim Fjord on Nov 3rd, 2014. The same transducer and hydrophone array as sea trial 2 were used. The hydrophone array was close to shore deployed with the hydrophone depth varied from 2 to 9 m. The transmitter was 1000 m away from the receiver array and the transducer was deployed at a depth of 15 m from Gunnerus. During the experiment, the communication signals were burst 3 times and one receiving hydrophone was connected with DSP board for online processing. Meanwhile, the received signals from all the hydrophones were also recorded by data acquisition device and retransmitted at lab. The achieved average BERs are presented in Table 8. For receiver with single hydrophone, the average BER after 4 iterations is below 10^{-3} and the worst BER performance during all received transmissions is less than 5×10^{-3} after 4 iterations. For receiver with two hydrophones, the average BER after 4 iteration was less than 10^{-4} and the worst BER performance after 4 iterations was less than 10^{-3} .

6. Conclusion

In this paper, we present our DSP based real-time underwater acoustic communication receiver. The receiver incorporates the modern iterative estimation and iterative equalization which could significantly improve system performance. Through careful optimization, our receiver system is implemented with large margin which could be used for future higher data receiving system. The performance of the developed system is assessed through both tank experiments and sea experiments and satisfied results are achieved. In current version, our receiver is relied on the assumption that the channel should satisfy the quasi-static condition i.e. channel can be regarded as time-invariant within one block. Future work can include the development of real-time receiver for both delay-spread and time-spread underwater acoustic channels.

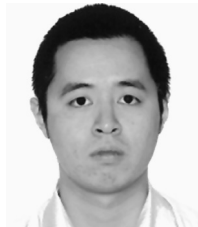
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References

- [1] D. Kilfoyle, A. Baggeroer, The state of the art in underwater acoustic telemetry, *IEEE J. Ocean. Eng.* 25 (1) (2000) 4–27. <http://dx.doi.org/10.1109/48.820733>.
- [2] B. Li, S. Zhou, M. Stojanovic, L. Freitag, P. Willett, Multicarrier communication over underwater acoustic channels with nonuniform doppler shifts, *IEEE J. Ocean. Eng.* 33 (2) (2008) 198–209. <http://dx.doi.org/10.1109/OE.2008.920471>.
- [3] B. Li, J. Huang, S. Zhou, K. Ball, M. Stojanovic, L. Freitag, P. Willett, MIMO-OFDM for high-rate underwater acoustic communications, *IEEE J. Ocean. Eng.* 34 (4) (2009) 634–644. <http://dx.doi.org/10.1109/OE.2009.2032005>.
- [4] Y. Zheng, C. Xiao, T.C. Yang, W.-B. Yang, Frequency-domain channel estimation and equalization for single carrier underwater acoustic communications, in: *OCEANS 2007, 2007*, pp. 1–6. <http://dx.doi.org/10.1109/OCEANS.2007.4449247>.
- [5] J. Zhang, Y. Zheng, C. Xiao, Frequency-domain equalization for single carrier mimo underwater acoustic communications, in: *OCEANS 2008, 2008*, pp. 1–6. <http://dx.doi.org/10.1109/OCEANS.2008.5151957>.
- [6] J.W. Choi, T. Riedl, K. Kim, A. Singer, J. Preisig, Adaptive linear turbo equalization over doubly selective channels, *IEEE J. Ocean. Eng.* 36 (4) (2011) 473–489. <http://dx.doi.org/10.1109/OE.2011.2158013>.
- [7] J. Tao, J. Wu, Y. Zheng, C. Xiao, Enhanced MIMO LMMSE turbo equalization: Algorithm, simulations, and undersea experimental results, *IEEE Trans. Signal Process.* 59 (8) (2011) 3813–3823. <http://dx.doi.org/10.1109/TSP.2011.2147782>.
- [8] J. Zhang, Y. Zheng, Frequency-domain turbo equalization with soft successive interference cancellation for single carrier mimo underwater acoustic communications, *IEEE Trans. Wireless Commun.* 10 (9) (2011) 2872–2882. <http://dx.doi.org/10.1109/TWC.2011.072511.100324>.
- [9] J. Huang, S. Zhou, Z. Wang, Performance results of two iterative receivers for distributed mimo ofdm with large doppler deviations, *IEEE J. Ocean. Eng.* 38 (2) (2013) 347–357. <http://dx.doi.org/10.1109/OE.2012.2223991>.
- [10] Bethos, Inc. Acoustic Modems, http://www.benthos.com/index.php/product_dashboard/acoustic_modems [Online].
- [11] DSPCOMM, http://www.dspcomm.com/products_aquacomm.html [Online].
- [12] LinkQuest, Inc. Underwater acoustic modems. http://www.linkquest.com/html/uwm_hr.pdf [Online].
- [13] L. Freitag, M. Grund, S. Singh, J. Partan, P. Koski, K. Ball, The whoi micro-modem: an acoustic communications and navigation system for multiple platforms, in: *OCEANS 2005, Vol. 2, 2005*, pp. 1086–1092. <http://dx.doi.org/10.1109/OCEANS.2005.1639901>.
- [14] H. Yan, L. Wan, S. Zhou, Z. Shi, J.-H. Cui, J. Huang, H. Zhou, DSP based receiver implementation for OFDM acoustic modems, *Phys. Commun.* 5 (1) (2012) 22–32. <http://dx.doi.org/10.1016/j.phycom.2011.09.001>.
- [15] T. Husoy, M. Pettersen, B. Nilsson, T. Oberg, N. Warakagoda, A. Lie, Implementation of an underwater acoustic modem with network capability, in: *OCEANS 2011, 2011*, pp. 1–10. <http://dx.doi.org/10.1109/Oceans-Spain.2011.6003642>.
- [16] B. Benson, Y. Li, B. Faunce, K. Domond, D. Kimball, C. Schurgers, R. Kastner, Design of a low-cost underwater acoustic modem, *IEEE Embedded Syst. Lett.* 2 (3) (2010) 58–61. <http://dx.doi.org/10.1109/LES.2010.2050191>.
- [17] R. Le Bidan, C. Laot, D. Leroux, Real-time MMSE turbo-equalization on the tms320c5509 fixed-point DSP, in: *ICASSP 2004, Vol. 5, 2004*, pp. V-325–V-328. <http://dx.doi.org/10.1109/ICASSP.2004.1327113>.
- [18] M. Morelli, U. Mengali, Carrier-frequency estimation for transmissions over selective channels, *IEEE Trans. Commun.* 48 (9) (2000) 1580–1589. <http://dx.doi.org/10.1109/26.870025>.
- [19] H. Liu, P. Schnitter, Iterative frequency-domain channel estimation and equalization for single-carrier transmissions without cyclic-prefix, *IEEE Trans. Wireless Commun.* 7 (10) (2008) 3686–3691. <http://dx.doi.org/10.1109/TWC.2008.070371>.
- [20] M. Tüchler, R. Koetter, A. Singer, Turbo equalization: principles and new results, *IEEE Trans. Commun.* 50 (5) (2002) 754–767. <http://dx.doi.org/10.1109/TCOMM.2002.1006557>.
- [21] Y. Wu, X. Zhu, A. Nandi, Low complexity adaptive turbo frequency-domain channel estimation for single-carrier multi-user detection, *IEEE Trans. Wireless Commun.* 7 (11) (2008) 4094–4099. <http://dx.doi.org/10.1109/TWC.2008.070570>.
- [22] M. Sabbaghian, D. Falconer, Joint turbo frequency domain equalization and carrier synchronization, *IEEE Trans. Wireless Commun.* 7 (1) (2008) 204–212. <http://dx.doi.org/10.1109/TWC.2008.060451>.

- [23] M. Morelli, L. Sanguinetti, U. Mengali, Channel estimation for adaptive frequency-domain equalization, *IEEE Trans. Wireless Commun.* 4 (5) (2005) 2508–2518. <http://dx.doi.org/10.1109/TWC.2005.853896>.
- [24] P. Yin, Introduction to TMS320C6000 DSP optimization, Texas Instruments Application Report.
- [25] A.V. Oppenheim, R.W. Schaffer, J.R. Buck, *Discrete-Time Signal Processing*, second ed., Prentice-Hall, Inc., Upper Saddle River, NJ, USA, 1999.



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